# PROJECT SUMMARY

**TalkToJesus — A Bilingual Voice-First AI Spiritual Companion**

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https://github.com/manish-gitx/rn-final-manish

*Prepared so that the viva examiner can understand the project before the session. The full argument, evidence and limitations are in the final project report.*

## 1. The problem

Telugu-language Bible applications have been downloaded by millions of users. They are well made, and they all share one property: content flows one way. A believer can read a passage, follow a plan or play a recorded hymn, but cannot ask what any of it means for the decision in front of them tonight.

Four gaps follow. Communication is one-way. Pastoral conversation requires another person to be available, scheduled and willing to be approached. Guidance is not aware of the asker's situation. And general-purpose AI assistants, which can hold a conversation, are neither grounded in scripture nor idiomatic in Telugu — addressing a believer as నా బిడ్డ is not something a literal translation gets right.

## 2. Unique Value Proposition

> **Every existing option covers some of the four. None covers all four at once.**
>
> **TalkToJesus is conversational, scripture-grounded, regional-language and voice-first simultaneously — and it is the only one of them that can tell you, per turn, where its seconds go.**

Set against the alternatives surveyed in Phase 1:

| System | Conversational | Scripture-grounded | Telugu | Voice-first |
|---|---|---|---|---|
| YouVersion Bible App | No | Yes | Partial | No |

| System | Conversational | Scripture-grounded | Telugu | Voice-first |
|---|---|---|---|---|
| Telugu Bible applications | No | Yes | Yes | No |
| Pray.com | No | Yes | No | Partial |
| General AI assistants | Yes | No | Weak | Partial |
| Glorify / Abide | No | Yes | No | Partial |
| **TalkToJesus** | **Yes** | **Yes** | **Yes** | **Yes** |

The second half of the proposition is the engineering one and matters more for assessment than for a user. Every conversational turn records the duration of transcription, generation and speech synthesis separately, alongside the total. The system therefore measures itself. As Section 5 records, that instrumentation produced a result which contradicted the project's own earlier documentation — which is the point of building it.

## 3. The solution

A user opens the application, presses one control, speaks in English or Telugu, and hears a spoken reply written in the first person, addressed as "My child" or "నా బిడ్డ", citing scripture by book, chapter and verse, and closing with a short prayer.

Around that: a bilingual Bible reader with six translations including two Telugu ones and an offline cache; twelve public-domain hymns bundled to play with no network; a private transcript of previous conversations; and a ₹499/month subscription that lifts a three-conversation free tier.

## 4. Architecture in brief

Three tiers. A **Flutter** client for iOS and Android using Riverpod. A stateless **Express 5 / TypeScript** API of 28 endpoints on **Google Cloud Run**, scaling from zero. **Supabase PostgreSQL** across eight tables, plus **OpenAI**, **ElevenLabs** and **Razorpay**.

A voice turn is three timed stages: **Whisper** transcribes, **GPT-4o** generates under a language-specific persona prompt with the previous six turns supplied as context, and **ElevenLabs** synthesises the reply as speech.

Supporting systems: Google OAuth exchanged for a signed token; Razorpay subscriptions reconciled by HMAC-SHA256-verified webhooks, with failed verifications deliberately recorded as evidence that checking happens; five runtime feature flags; and a dependency-free administrative console with live metrics, content management, a webhook audit and an audit trail.

**A deliberate clarification.** There is no retrieval-augmented generation, no embedding and no vector index. Context is the previous turns replayed into the model's message array. Scripture citations come from the model's parameters, not from a retrieved corpus, and their accuracy was not formally verified.

## 5. Results

**All 20 functional requirements are implemented**; one (the audio player's previous/next controls) is partially met.

**141 automated tests pass** — 66 backend across 10 Jest suites, 75 client across 9 Flutter files, none failing, none skipped. Both suites were re-run for the report and the output captured as evidence.

**The full subscription and conversation flow was verified end to end on physical hardware**, including a real UPI payment and the resulting webhook reconciliation.

**The measured latency fails its requirement, and finding that is the project's most useful result.** NFR1 specifies a five-second pipeline. The live instance measures:

| Stage | p50 | Share |
|---|---|---|
| Speech to text (Whisper) | 4,004 ms | 15 % |
| AI response (GPT-4o) | 3,471 ms | 13 % |
| Text to speech (ElevenLabs) | **19,618 ms** | **71 %** |
| **End to end** | **27,457 ms** | |

Sample: two voice turns — enough to establish an order of magnitude, not a distribution.

The three-to-six second figure quoted in the Phase 3 document and in the presentation deck is traceable to the administrative console's *demonstration fixtures* (3,030 ms), which are hand-written constants, not a measurement. The report records this correction in full.

The distribution is the useful part: latency is a speech-synthesis problem, not a model problem. And because the persona prompt specifies a two-to-four sentence reply while the model actually returns three paragraphs, the most probable single cause is the prompt rather than the vendor — the system is asking a speech engine to render four to five times more text than designed.

## 6. Honest limitations

Recorded in full in the report; the material ones are: latency fails NFR1 by roughly a factor of five; row-level security is enabled but unpolicied and bypassed by the API's service key, so authorization rests on application code with no test asserting isolation between users; a long-lived token and two analytics keys are committed to source control and must be rotated; there is no rate limiting and CORS is open; conversation transcripts are personal data stored indefinitely with no retention or deletion policy; test coverage stops at the service layer with no widget, integration or end-to-end tests; and three bundled hymns are CC BY-SA, which requires a credits screen the application does not yet have.

## 7. Demonstration

| Recording | Duration | Link |
|---|---|---|
| Application | 3 min 50 s | https://drive.google.com/file/d/124faeA_zm2HM7rPP_rWwJPsXSj6iN4QS/view |
| Administrative console | 1 min 13 s | https://drive.google.com/file/d/1UeTyKBB8uDYdNhWjS14s2lbwYxO7xW6p/view |

The console recording shows demonstration mode for its first fifty seconds and the live instance thereafter — the two halves report figures differing by an order of magnitude, for the reason given in Section 5.